

Clausius on the Cut: the CHM Envelope Beats a Flat Kernel on the Horizon, Not in the Bulk

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Abstract

Paper 26 scored every hop. A shapeless local kernel won. Jacobson puts the first law on the *horizon*. This note splits the ball.

Bulk ($r_{\text{mid}} \leq R/2$, $N = 12$, $R = 4$, 84 hops): flat still wins. The envelope is not selected inside.

Surface ($r_{\text{mid}} \geq 0.75R$) on a *new* grid $N = 10$, $R = 3$, 150 hops, same C3: $R_{\text{CHM}} = 0.919 < R_{\text{flat}} = 0.996$. Auditor $N = 12$, $R = 3$: $0.818 < 0.969$. C3 PASS, CONFIRMED.

Tracking floors fail ($|\rho| = 0.39$ and 0.58 , below 0.60). This is not a strong first-law theorem. It is the first time the geometric weight wins the effective null, and it wins where the horizon is.

Not $\eta = 1/4G$. Not Einstein. Not de Sitter.

1 Why the split

A single-hop δS is largest at the cut (Paper 26 adversary). Mixing bulk and surface lets a flat kernel ride the area-law surface. Scoring the cut alone is the Clausius probe. The bulk run was locked first and lost. The surface run uses a different N so the $N = 12$, $R = 4$ surface numbers are not promoted.

2 Numbers

	n	R_{CHM}	R_{flat}	C3
bulk $N = 12, R = 4$	84	0.904	0.868	flat
surface $N = 10, R = 3$	150	0.919	0.996	CHM
surface audit $N = 12, R = 3$	150	0.818	0.969	CHM

The geometric weight is doing work on the cut. The correlation is still weak. Road A now has: local K (A2), area law (A1), CHM-*type* shape of K (Paper 25), and a horizon Clausius that prefers that shape to a flat kernel (C3 only).

Admission. *I did not move C3 onto the surface of the same sample that suggested it. I did not lower the tracking floors after they failed. I did not write that Einstein has been derived.*